

Zeeman effect

In a magnetic field $B = |\mathbf{B}|$ we have for hydrogen atoms

$$E_{nljm} = -\frac{\alpha^2 \mu c^2}{2n^2} \left[1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) \right] + g_J m_j \mu_B B$$

where

$$l = 0, 1, 2, \dots, n-1$$

and

$$j = |l \pm \frac{1}{2}|, \quad m_j = \pm \frac{1}{2}, \pm \frac{3}{2}, \dots, \pm j$$

Symbol g_J is the Landé g -factor

$$g_J = 1 + \frac{j(j+1) - l(l+1) + \frac{3}{4}}{2j(j+1)}$$

For $n = 2$ there are eight energy levels.

n	l	j	m_j	E_{nljm}
2	1	$\frac{3}{2}$	$\frac{3}{2}$	$-\frac{\alpha^2 \mu c^2}{8} (1 + \frac{1}{16} \alpha^2) + 2\mu_B B$
2	1	$\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{\alpha^2 \mu c^2}{8} (1 + \frac{1}{16} \alpha^2) - 2\mu_B B$
2	1	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{\alpha^2 \mu c^2}{8} (1 + \frac{1}{16} \alpha^2) + \frac{2}{3} \mu_B B$
2	1	$\frac{3}{2}$	$-\frac{1}{2}$	$-\frac{\alpha^2 \mu c^2}{8} (1 + \frac{1}{16} \alpha^2) - \frac{2}{3} \mu_B B$
2	1	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{\alpha^2 \mu c^2}{8} (1 + \frac{5}{16} \alpha^2) + \frac{1}{3} \mu_B B$
2	1	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{\alpha^2 \mu c^2}{8} (1 + \frac{5}{16} \alpha^2) - \frac{1}{3} \mu_B B$
2	0	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{\alpha^2 \mu c^2}{8} (1 + \frac{5}{16} \alpha^2) + \mu_B B$
2	0	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{\alpha^2 \mu c^2}{8} (1 + \frac{5}{16} \alpha^2) - \mu_B B$

Eigenmath script